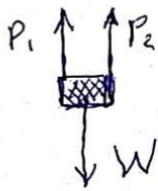


1 | Considerando el modelo de la partícula material, sin tener en cuenta las dimensiones de la viga, el peso de la viga se distribuye de manera uniforme a las fuerzas aplicadas por las dos personas.



$$P_1 + P_2 = W$$

$$P + P = W$$

$$2P = W$$

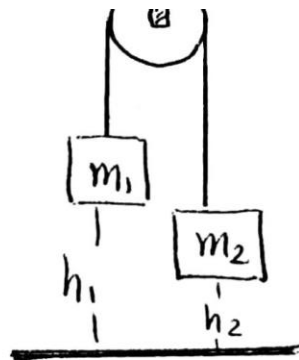
$$P = \frac{400}{2}$$

$$P = 200 \text{ N}$$

$$P_1 = P_2 = P$$

Por tanto,
es independi-
ente del
ángulo.

F2



La energía
mecánica del
sistema se
conserva.

$$E_i = E_f$$

$$m_1 g h_1 + m_2 g h_2 = m_1 g (h_1 + h_2) + \frac{1}{2} (m_1 + m_2) v^2$$

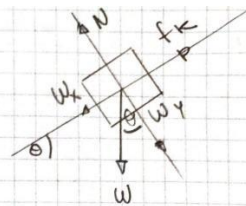
$$4(9,8)h_1 + (6)(9,8)(3) = (4)(9,8)(h_1 + 3) + \frac{1}{2}(10)v^2$$

$$176,4 = 117,6 + 5v^2$$

$$v = 3,4 \text{ m/s} //$$

F3

(8) $\uparrow \sum F_y = 0$
 $N - W_y = 0$
 $N = mg \cos \theta \dots (1)$



$$W_N = W_{W_x} + W_{f_k}$$

$$W_N = W_x d \cos \theta + f_k d \cos \theta$$

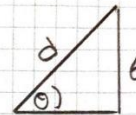
$$W_N = mg \sin \theta d - \mu_k mg \cos \theta d$$

$$W_N = mg \left(\sin \theta \frac{6}{\sin \theta} - \mu_k \cos \theta \left(\frac{6}{\sin \theta} \right) \right)$$

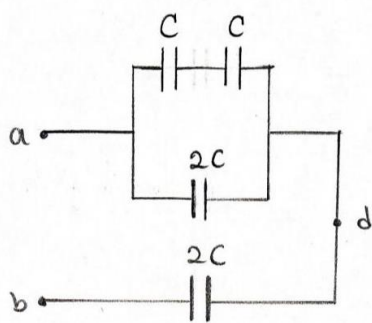
$$W_N = 2(10) \left[6 - \frac{1}{2} \left(\frac{4}{3} \right) \frac{6}{\frac{3}{5}} \right]$$

$$W_N = 20 [6 - 4]$$

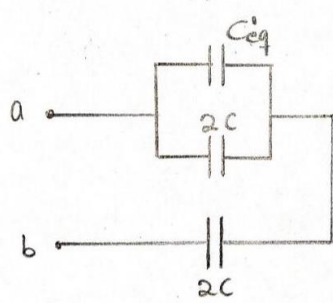
$$W_N = 40 \text{ J}$$



F4



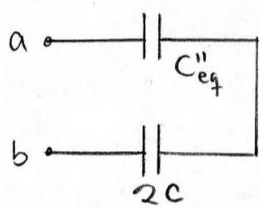
Datos:
 $C = 9 \mu f$



$$\frac{1}{C'_{eq}} = \frac{1}{C} + \frac{1}{C}$$

$$\frac{1}{C'_{eq}} = \frac{2}{C}$$

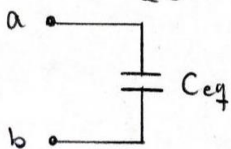
$$C'_{eq} = \frac{C}{2}$$



$$C''_{eq} = C'_{eq} + 2C$$

$$C''_{eq} = \frac{C}{2} + 2C$$

$$C''_{eq} = \frac{5C}{2}$$



$$\frac{1}{C_{eq}} = \frac{1}{C''_{eq}} + \frac{1}{2C}$$

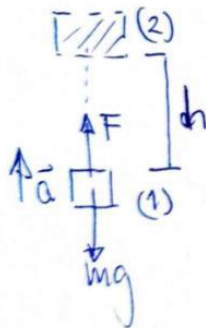
$$\frac{1}{C_{eq}} = \frac{2}{5C} + \frac{1}{2C} =$$

$$= \frac{4+5}{10C} = \frac{9}{10C}$$

$$C_{eq} = \frac{10C}{9} = 10 \mu f$$

(d)

P-3



$$\uparrow \Sigma F_y = ma$$

$$F - mg = ma$$

$$160 - 15 \cdot 10 = 15a$$

$$10 = 15a$$

$$a = \frac{2}{3} \text{ m/s}^2$$

$$v_2^2 = v_1^2 + 2ah$$

$$4 = 2 \cdot \frac{2}{3} h$$

$$\underline{h = 3 \text{ m}}$$

$$W = F \Delta y \cos \theta \quad \theta = 0$$

$$W = Fh = 160 \cdot 3$$

$$\underline{W = 480 \text{ J}} \quad \text{Sol.}$$